Machine Learning

AIML CZG565/ DSECLZG565

M1 : Introduction

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Contact Session 1

Agenda

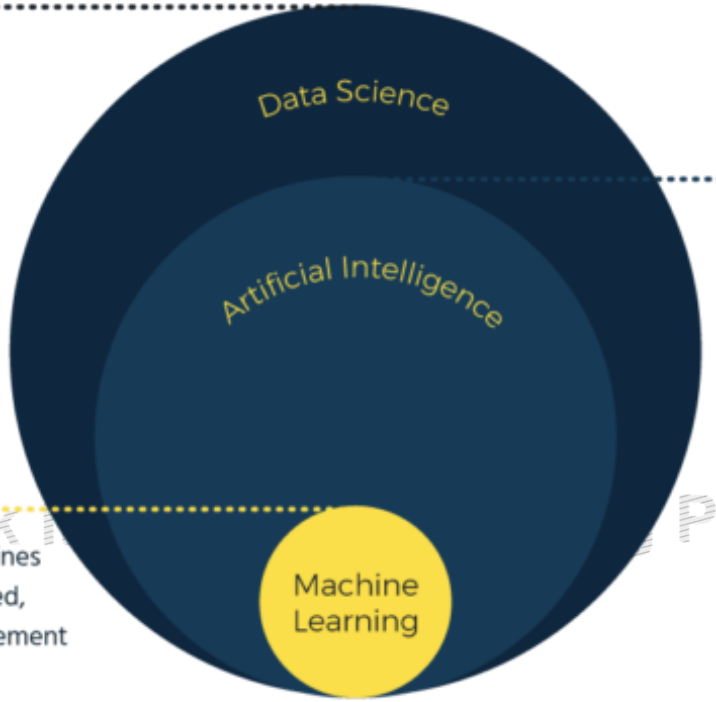
- What is Machine Learning?
- Why Machine Learning is important?
- Types of Machine Learning
- Application Areas
- Issues in Machine Learning
- Demo Case study

Work Integrated Learning Programmes

AI vs. Data Science vs. Machine Learning

Data Science

- Collection, preparation, and analysis of data
- Leverages AI/ML, research, industry expertise, and statistics to make business decisions



Artificial Intelligence

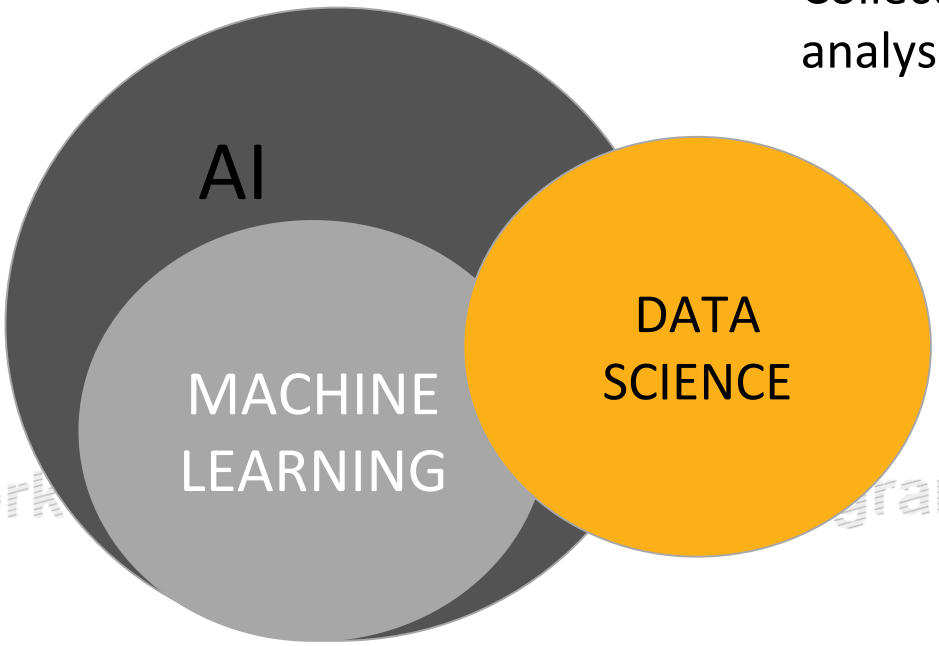
- Technology for machines to understand/interpret, learn, and make 'intelligent' decisions
- Includes Machine Learning among many other fields

Machine Learning

- Algorithms that help machines improve through supervised, unsupervised, and reinforcement learning
- Subset of AI and Data Science tool

AI vs DS vs ML

Collection, preparation and analysis of data

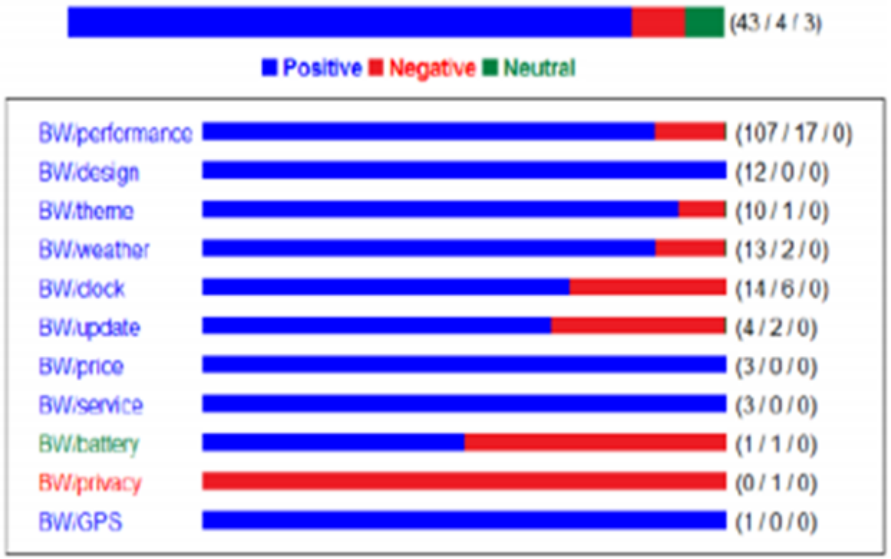


Common Use Cases

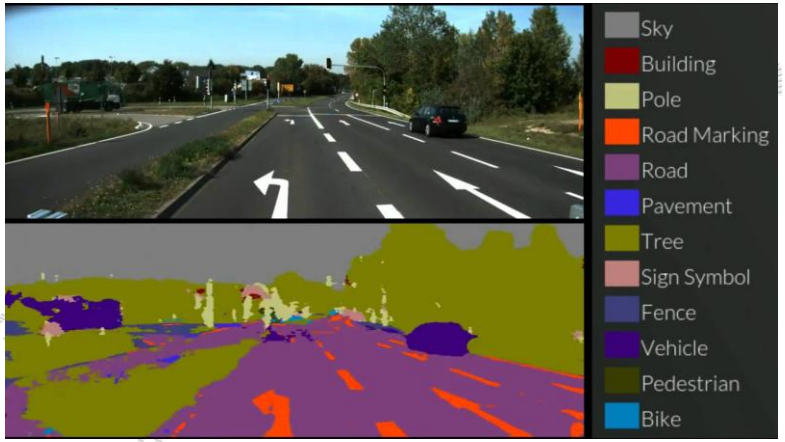
1. Applications in Smartphones
2. Transportation Optimization
3. Popular Web Services
4. Sales and Marketing
5. Security
6. Financial Domain

Introduction to Machine Learning

Common Use cases - Security & Transaction Domain



Sentiment analysis on Product review of Mobile phone



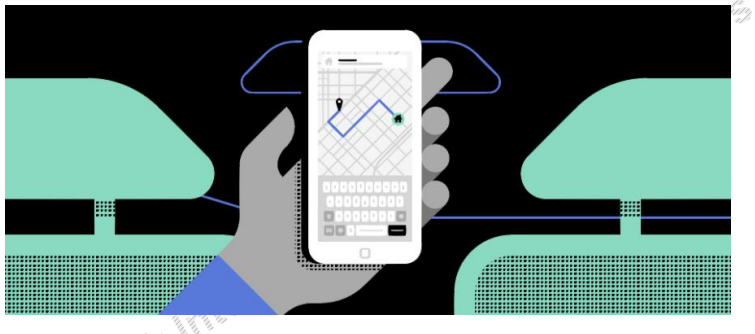
- Self-Driving Cars
- Fraud Detection in Banking
- Email Filtering
- Dynamic Pricing in Travel

Derived Applications:

> Cyber Security

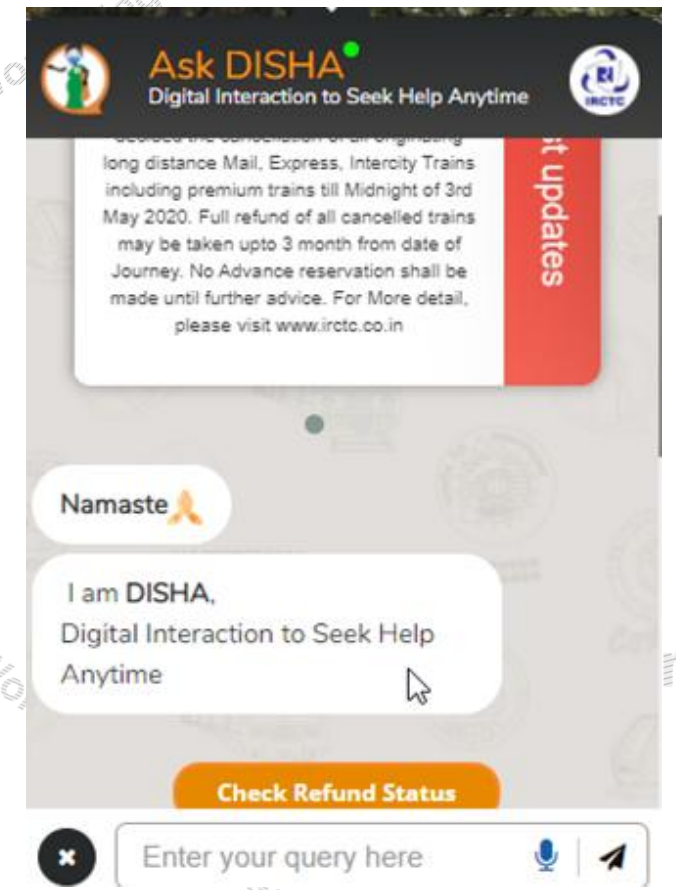
> Video Surveillance

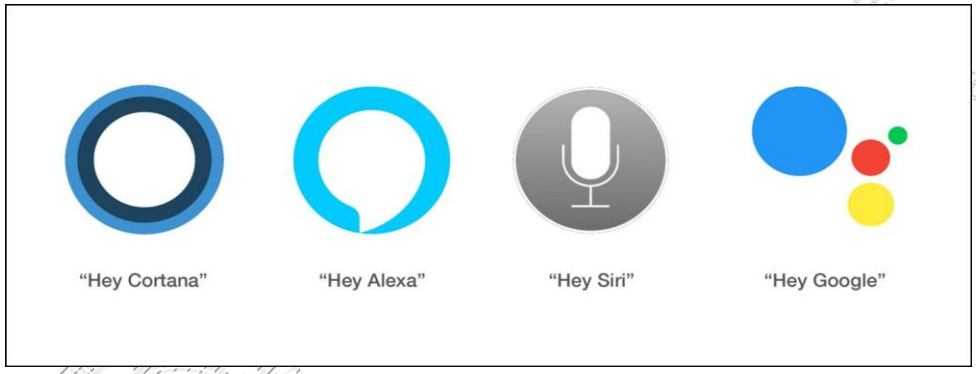
> Object Detection



Introduction to Machine Learning

Common Use cases - Customer Support Systems





- Apple's Siri
- Google Assistant
- Amazon's Alexa
- Google Duplex
- Microsoft's Cortana
- Samsung's Bixby

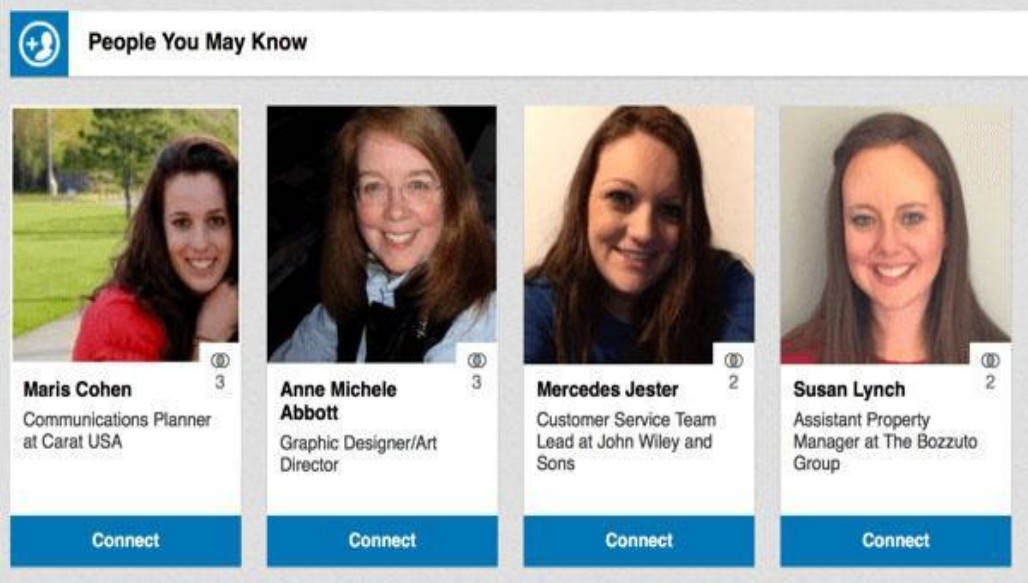
Derived Applications:

> Customer Support Query (Voice vs Text)

> Chatbots

Introduction to Machine Learning

Common Use cases - Recommendation Engines



- E-commerce sites like Amazon and Flipkart
- Book sites like Goodreads
- Movie services like IMDb and Netflix
- Hospitality sites like MakeMyTrip, Booking.com, etc.
- Retail services like StitchFix
- Food aggregators like Zomato and Uber Eats

Derived Applications:

> Personalized Marketing

> Personalized Banking

ML – What, When, Where ?

Work Integrated Learning Programmes

Introduction to Machine Learning

What is Machine Learning (ML)?

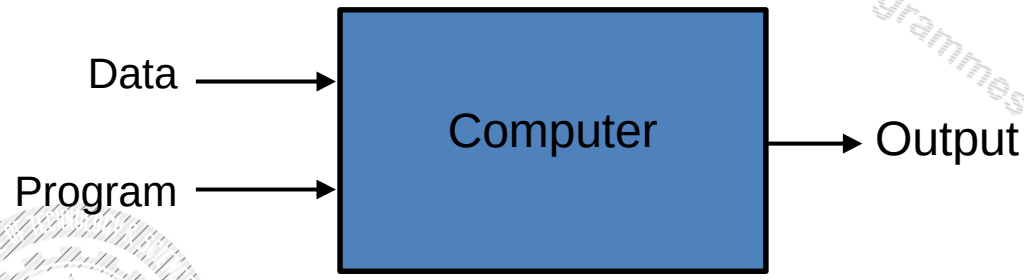
- The science (and art) of programming computers so they can *learn from data*
- More general definition

  Field of study that gives computers the ability to learn without being explicitly programmed
- Engineering-oriented definition

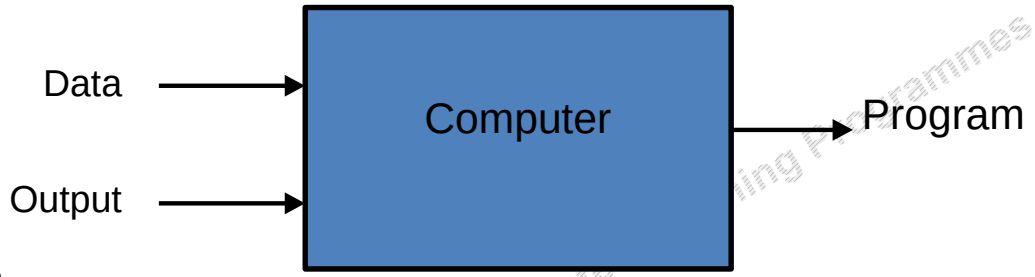
  Algorithms that improve their performance P at some task T with experience E

  A well-defined learning task is given by <*P*, *T*, *E*>

Traditional programming



Machine Learning



Slide credit: Pedro Domingos

Defining the Learning Tasks

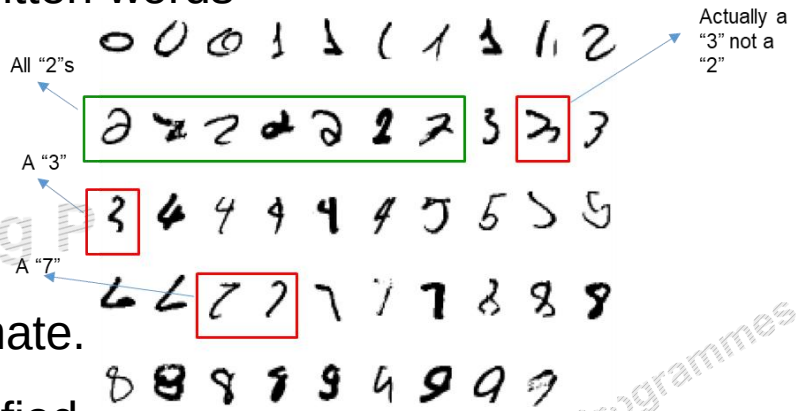
Improve on task T, with respect to performance metric P, based on experience E

Example 1

T: Recognizing hand-written words

P: Percentage of words correctly classified

E: Database of human labelled images of handwritten words



Example 2

T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

Defining the Learning Tasks

Improve on task T, with respect to performance metric P, based on experience E

Example 3

T: Playing Checkers

P: Percent of games won against opponents

E: Games Played against itself

Example 4

T: Drive on public four-lane highways using vision sensors.

P: Average distance travelled before an error (as judged by human).

E: A sequence of images and steering commands recorded while observing a human driver

Traditional Approach - Spam Filtering

Spam typically uses words or phrases such as “4U,” “credit card,” “free,” and “amazing”

Solution

Write a detection algorithm for frequently appearing patterns in spams

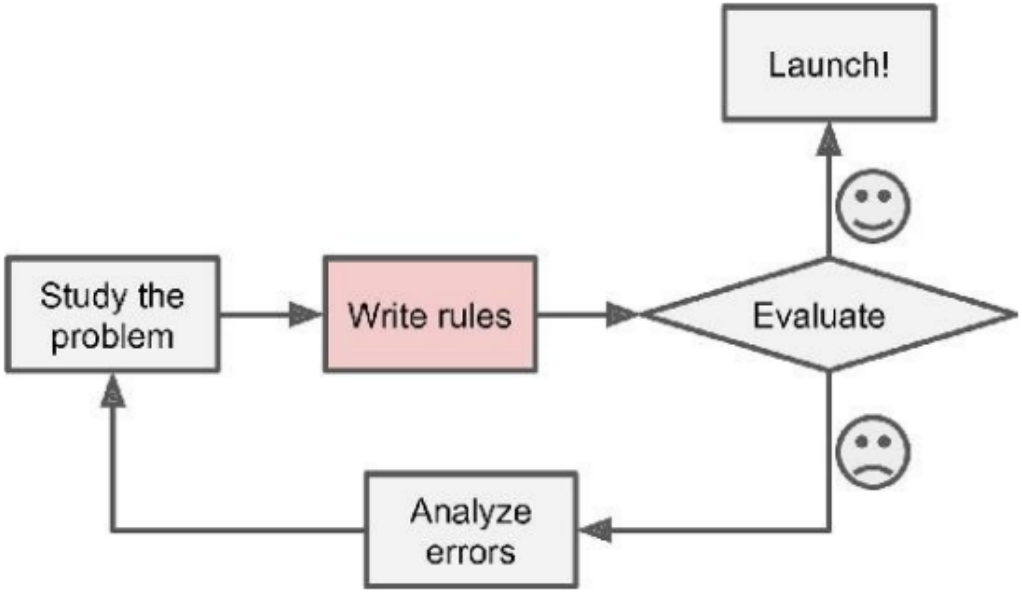
Test and update the detection rules until it is good enough.

Challenge

Detection algorithm likely

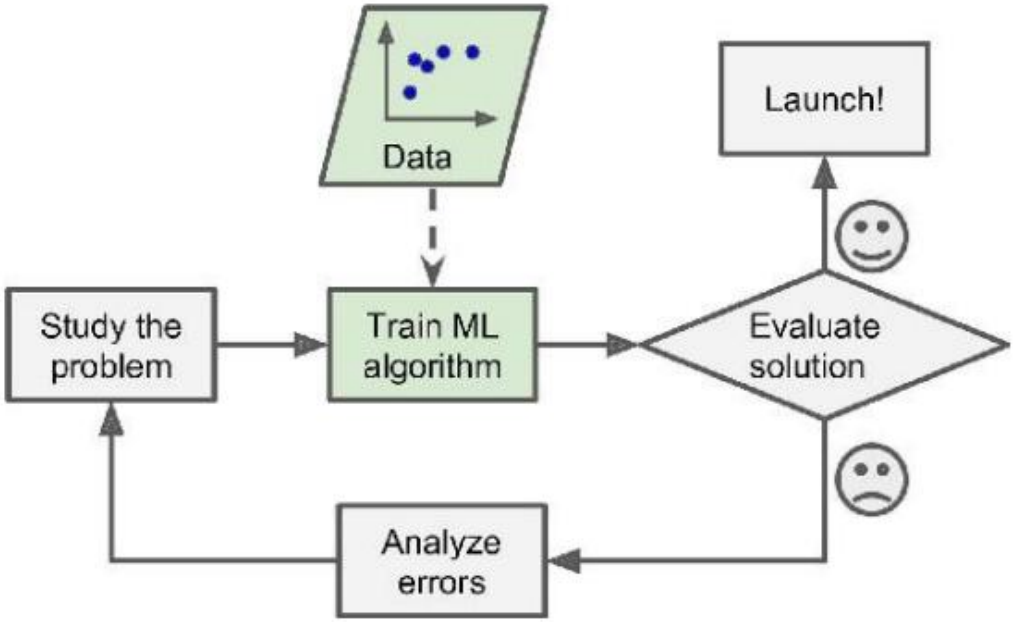
to be a long list of complex rules

hard to maintain.



ML Approach - Spam Filtering

Automatically learns phrases that are good predictors of spam by detecting unusually frequent patterns of words in spams compared to “ham”s



The program is much shorter, easier to maintain, and most likely more accurate.

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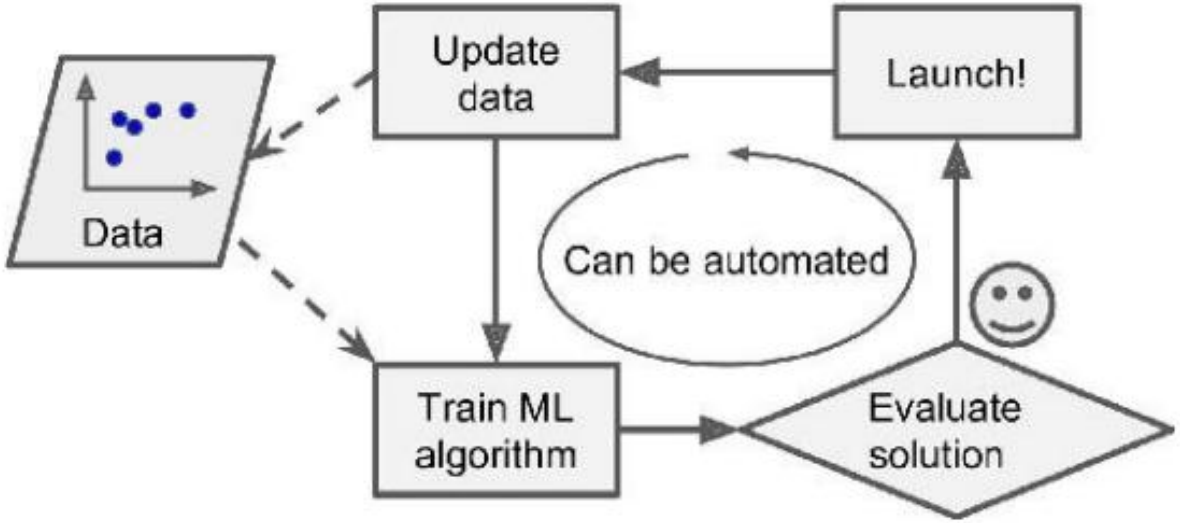

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Applications - Perspectives

Object Categorization Prediction

Sequence

Forecasting

Medical Research

Recommendation System

Content Management

Medical Diagnosis

Transaction Analysis

Recommendation System

Speech – Text Processing

Introduction to Machine Learning

Applications - Perspectives

Planning

Navigation

Path Finding

Gaming

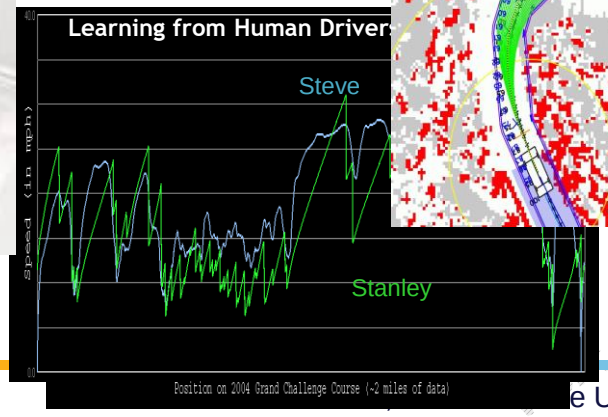
Controlling

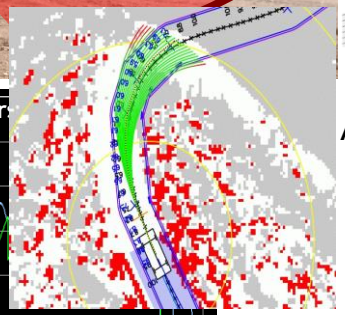
Problem Solving

Pattern Recognition

Optimization

Decision System





Adaptive Vision

Why ML

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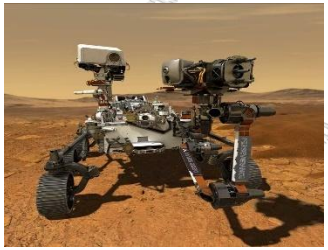
ML is used when:

- Human expertise does not exist (navigating on Mars)

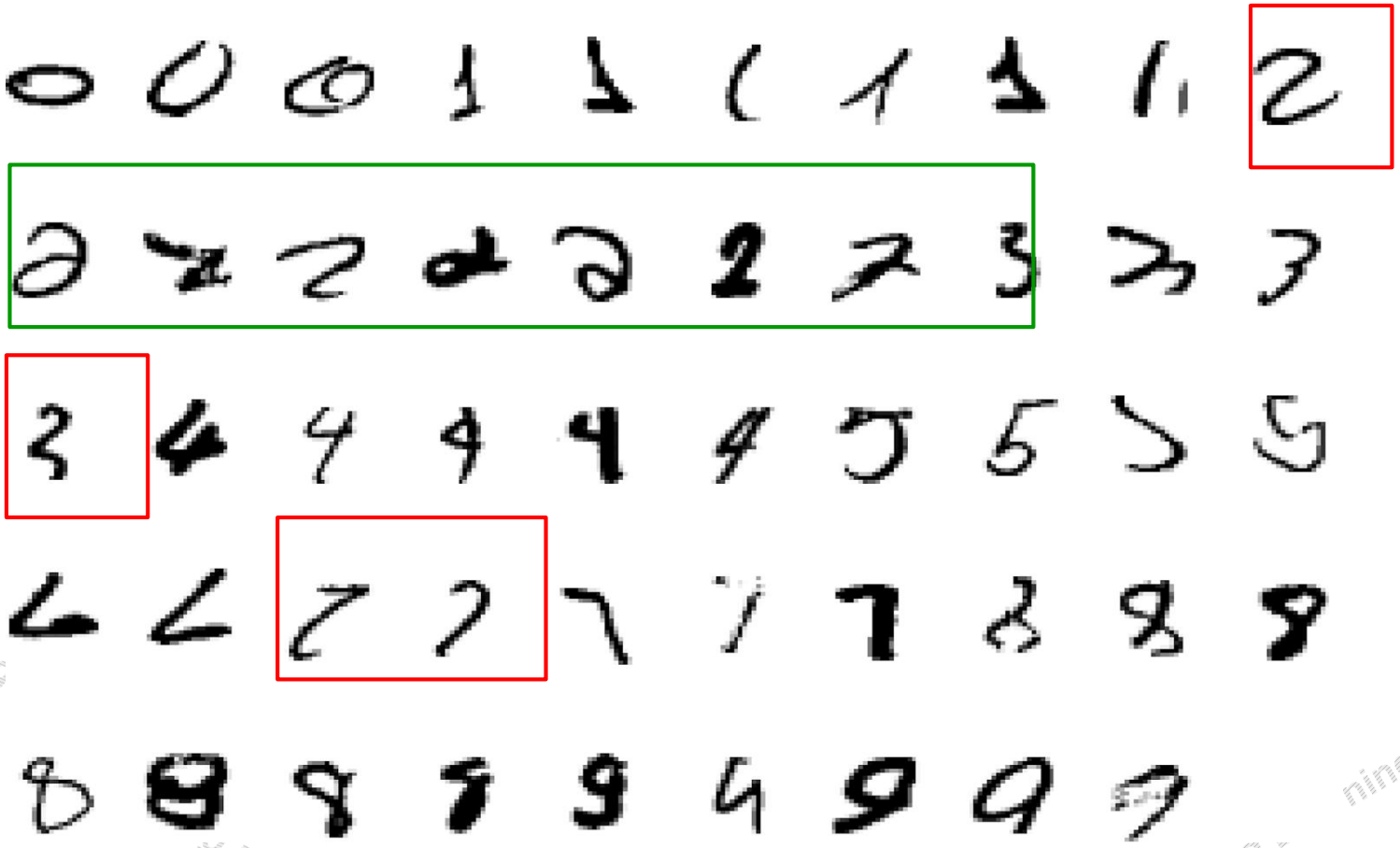


- Humans can’t explain their expertise (Biometrics)

- Models must be customized (personalized medicine)

- Learning isn’t always useful:
  - There is no need to “learn” to calculate payroll

Pattern recognition



Slide credit: Geoffrey Hinton

Types of ML

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Introduction to Machine Learning

Types : Inputs: Based on level of supervision

Feedback	No Feedback	Delayed Feedback (rewards/penalty)
Supervised	Unsupervised	Reinforcement

Machine Learning - Examples

Objective: Employability Prediction

Features / Attributes / Predictors

- ✓ CGPA
- ✓ Communication Skills
- ✓ Aptitude
- ✓ Programming Skills

S.No.	CGPA	Communication Skills	Aptitude	Programming Skills	Job Offered?
1	9.1	Average	Good	Excellent	Yes
2	8.4	Good	Good	Good	Yes
3	8.3	Poor	Average	Average	No
4	7.1	Average	Good	Average	No
5	8.2	Good	Excellent	Excellent	No

Machine Learning - Examples

Objective : Predicting price of a used car

Features / Attributes / Predictors

- ✓ Brand
- ✓ Year (Mfg)
- ✓ Engine Capacity
- ✓ Mileage
- ✓ Distance travelled
- ✓ Cab?

S.No	Brand	Year (Mfg)	Engine Capacity	Mileage	Distance travelled	Cab ?	Price (in Rs.)
1.	Honda City ZX	2008	1100	10.5	45000	N	3,50,000
2							
3							
4							

Machine Learning - Examples

Objective: Market Segmentation Study

Features / Attributes / Predictors

- ✓ Family income
- ✓ # of visits in a month
- ✓ Average money spent in a month
- ✓ Zip code

Customers for a retailer may fall into

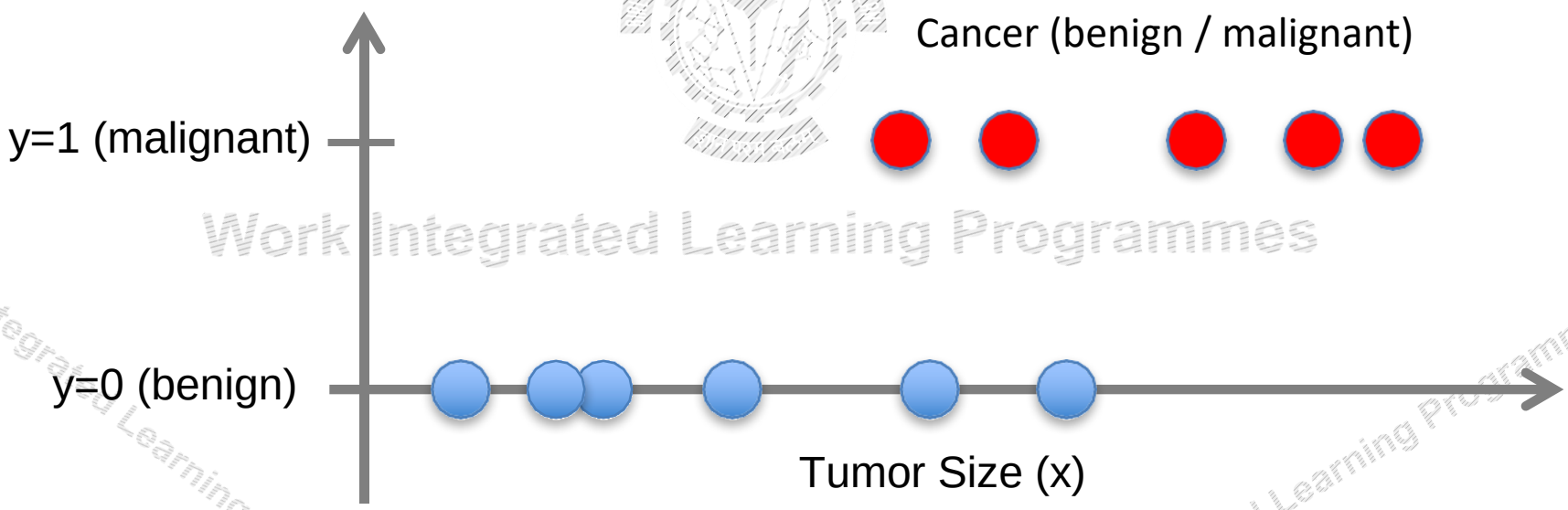
- ✓ two groups say big spenders and low spenders
- ✓ three groups say big spenders, medium spenders and low spenders
- ✓ Four groups, ….

S.No.	Zip Code	Family Income	# of visits in a month	Average Money Spent in a month
1	500078	11,50,000	4	8,000

Supervised Learning: Classification

GOAL : Previously unseen records should be assigned a class as accurately as possible.

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is categorical



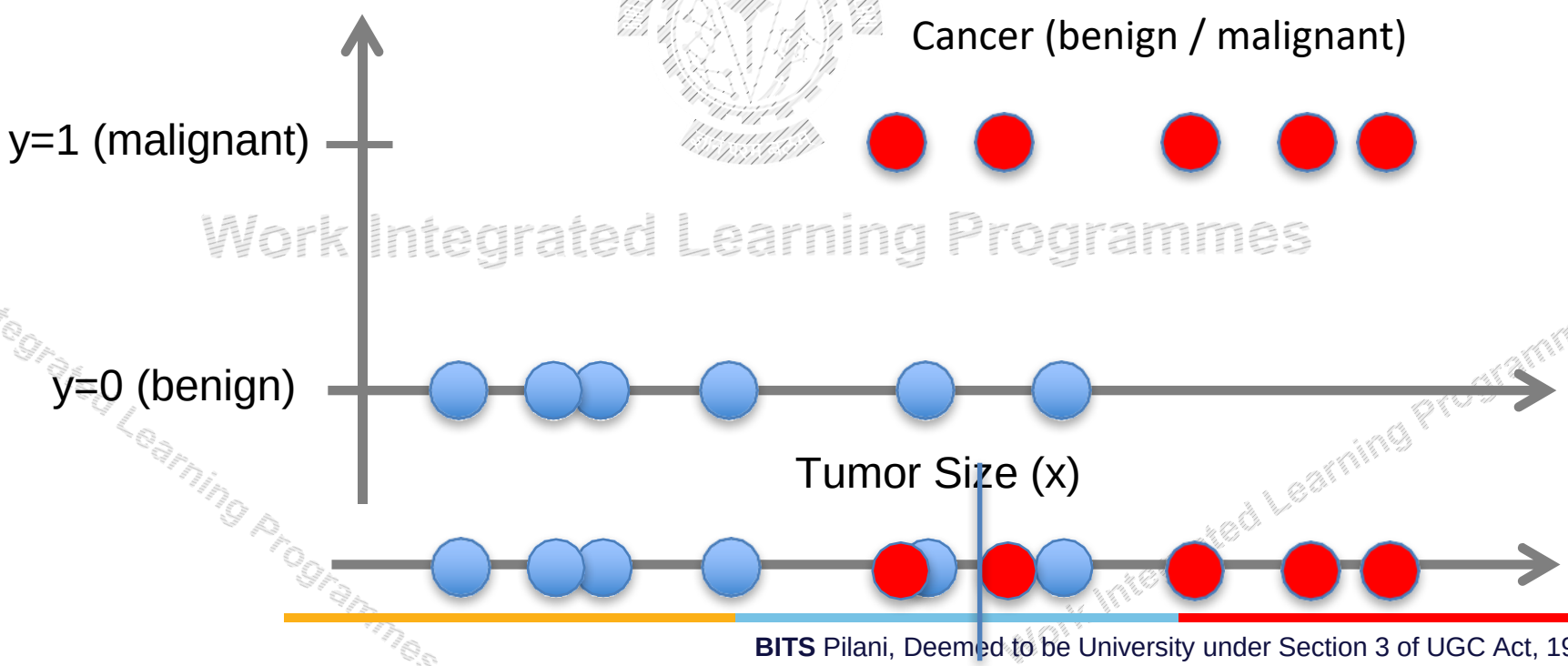
Slide Credit: Eric Eaton

Introduction to Machine Learning

Supervised Learning: Classification

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is categorical

Learnt classifier
If x>T, malignant else benign



Introduction to Machine Learning

Classification Applications

- Google Image Classification
- Face recognition system
- Spam filters – Specific Controls
- Document tagging
- Fraud detection

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Introduction to Machine Learning

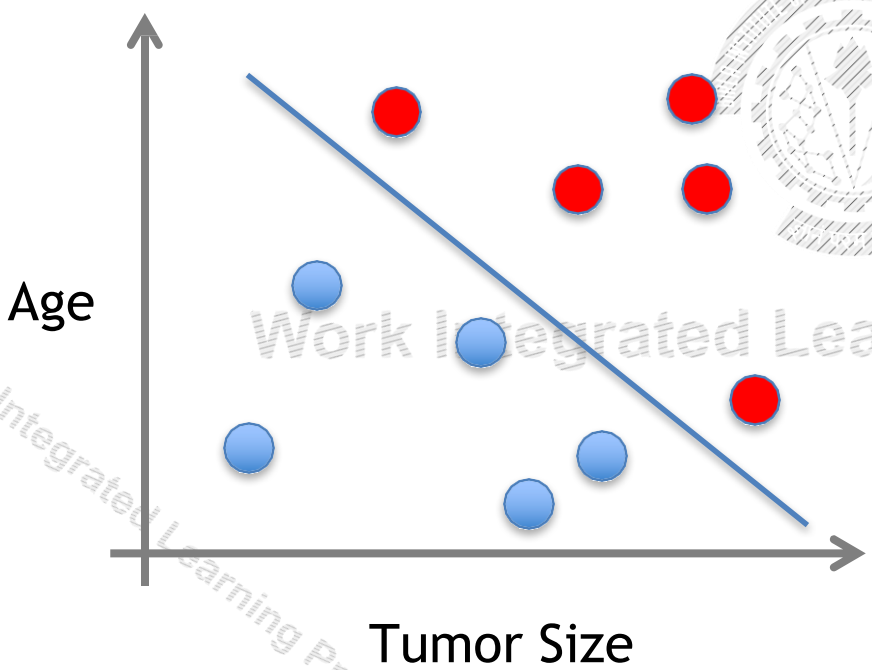
Supervised Learning: Classification

- *x* can be multi-dimensional
  - Each dimension corresponds to an attribute

Increasing Feature Dimension

- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape

…



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Supervised Learning Techniques / Algorithms

- Linear Regression
- Logistic Regression
- Naïve Bayes Classifiers
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural networks

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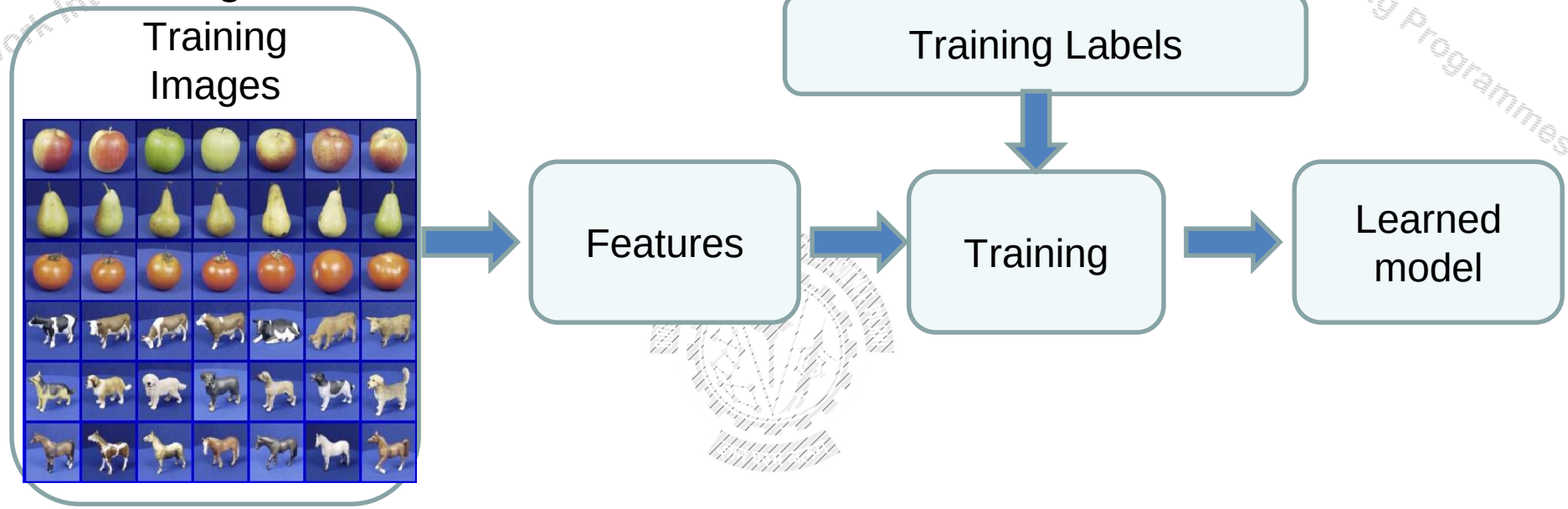
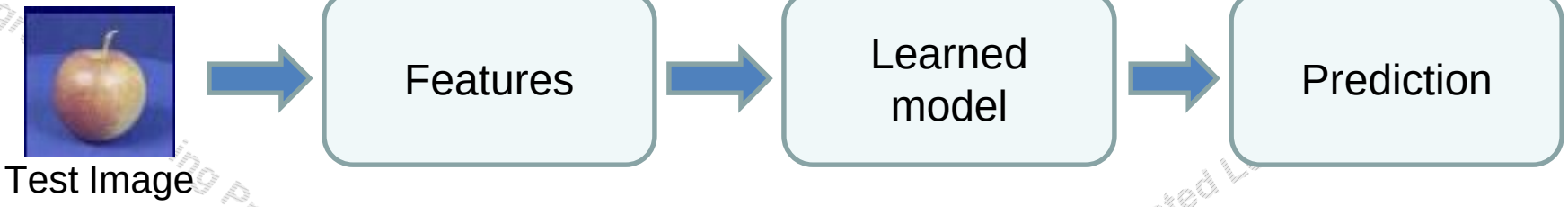
Training



Testing



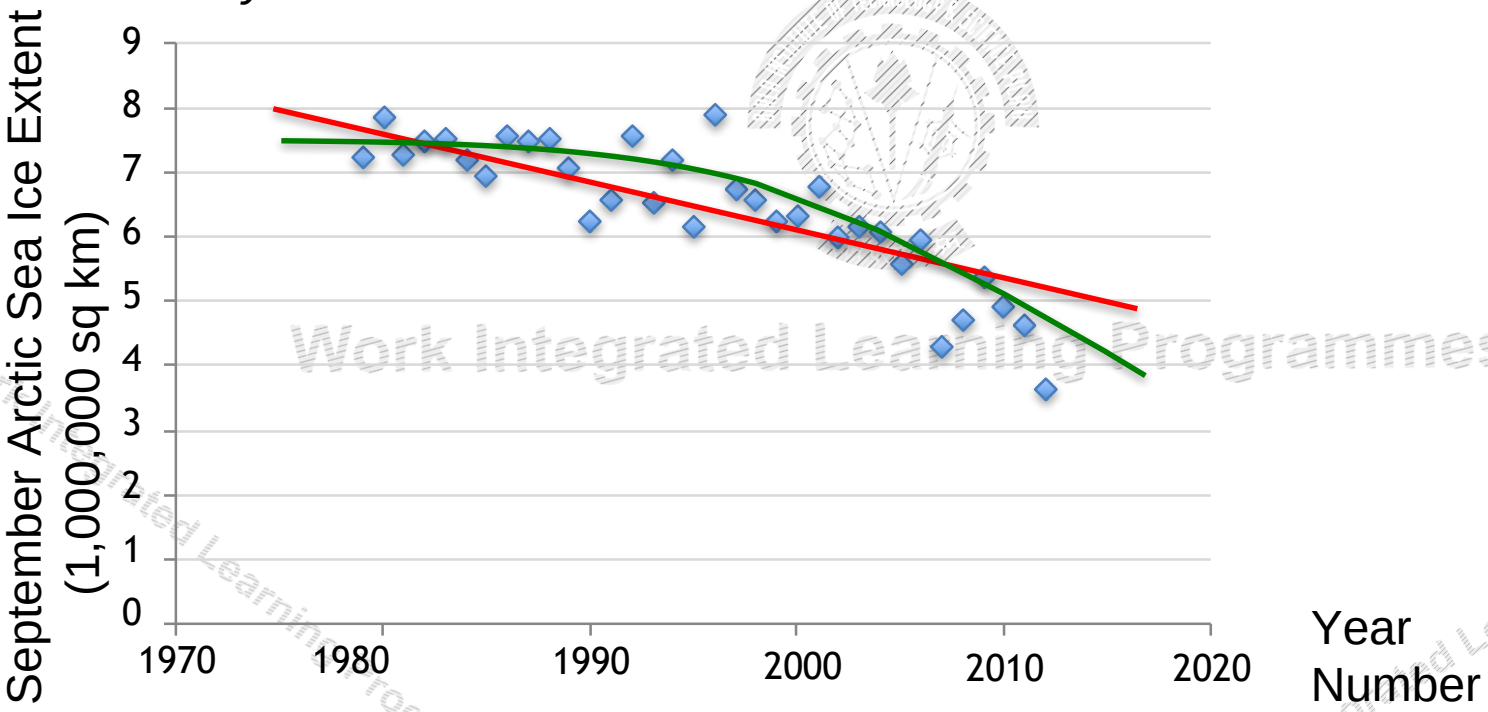
Slide credit: D. Hoiem and L. Lazebnik

Introduction to Machine Learning

Supervised Learning: Regression

GOAL : Previously unseen records should be assigned a value as accurately as possible.

- Given $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
  - y is real-valued

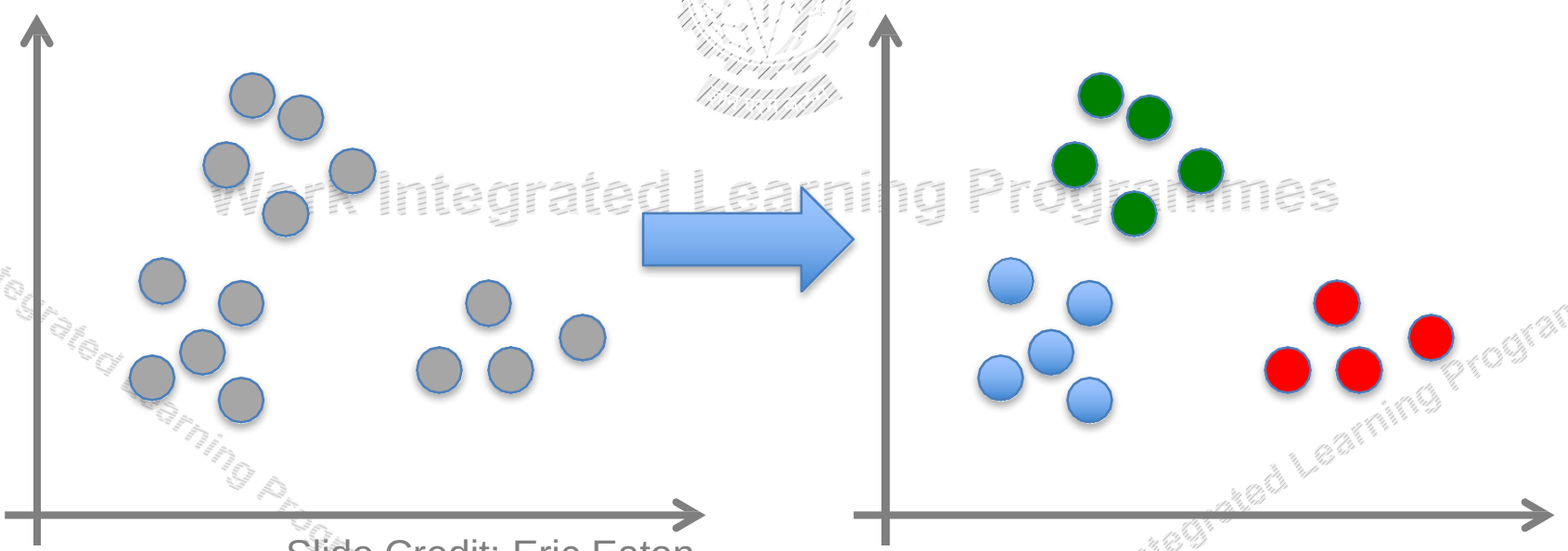


Data from G. Witt. Journal of Statistics Education, Volume 21, Number 1 (2013) Slide Credit: Eric Eaton

Unsupervised Learning

GOAL : Intra cluster distances are minimized and inter cluster distances are maximized

- Given $x_1, x_2, ..., x_n$ (without labels)
- Output hidden structure behind the x's
  - e.g., clustering

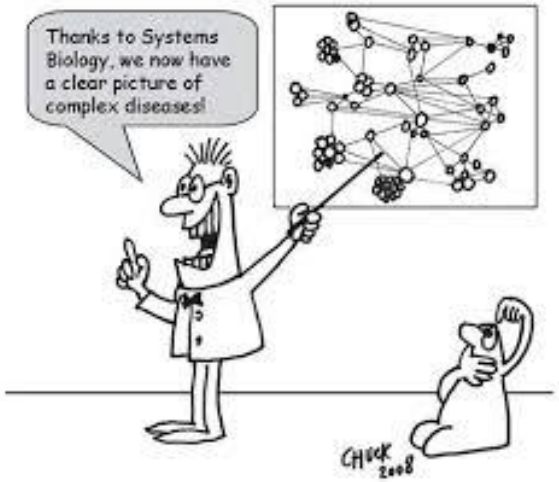


Slide Credit: Eric Eaton

Introduction to Machine Learning

Unsupervised Learning Applications

- Personalized recommendation system
- Targeted marketing
- Spam Filters
- Content Management – News hosted in Web
- Campaigning



Work Integrated Learning Programmes

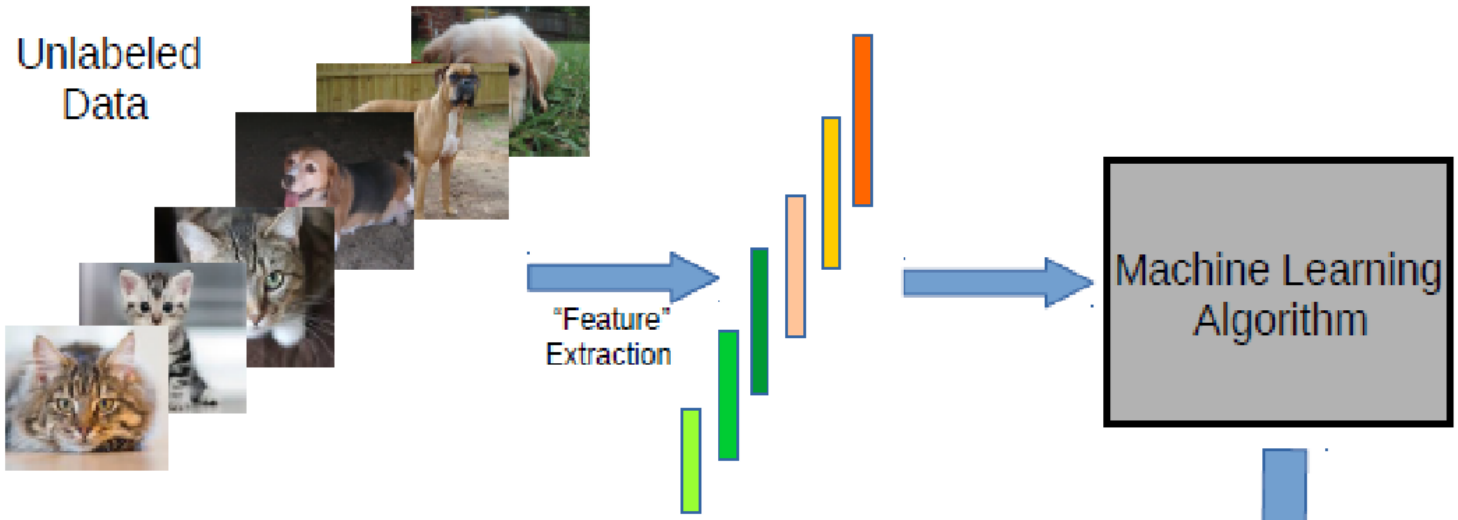
Unsupervised Learning Techniques

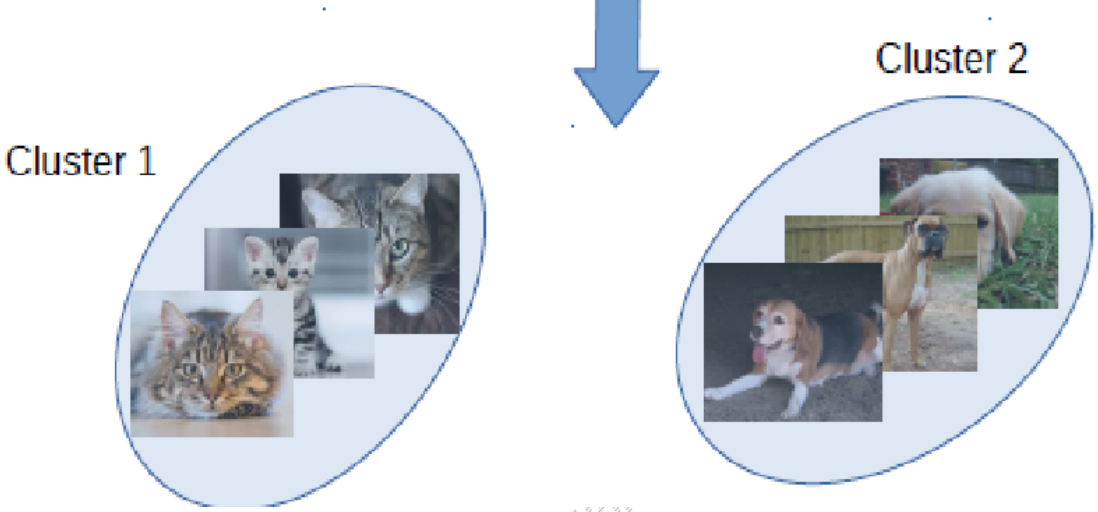
Clustering

- k-Means
- Hierarchical Cluster Analysis
- Expectation Maximization

Visualization and dimensionality reduction

- Principal Component Analysis (PCA)
- Kernel PCA
- Locally-Linear Embedding (LLE)
- t-distributed Stochastic Neighbor Embedding (t-SNE)

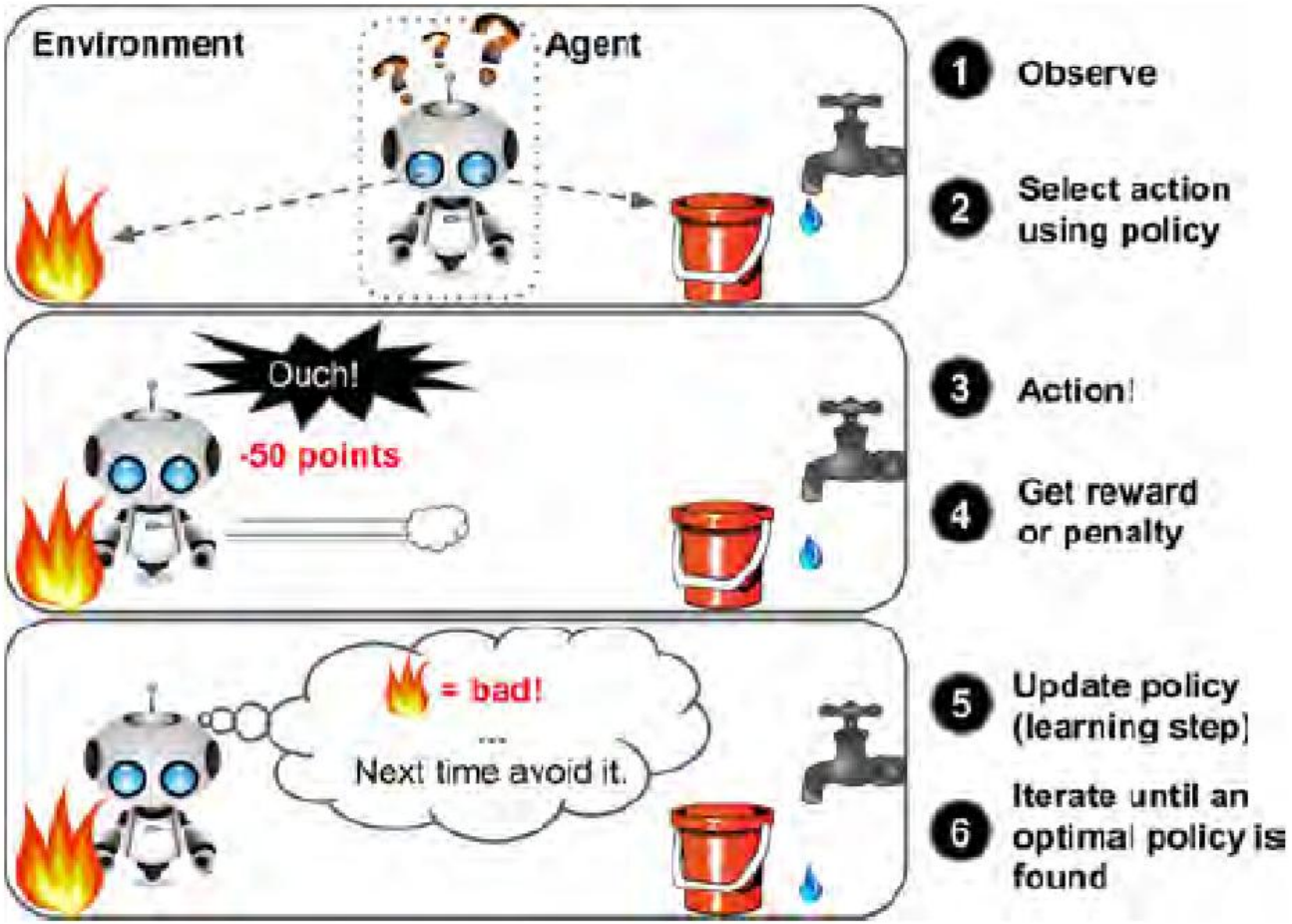




Note: Unsupervised Learning too can have (and often has) a "test" phase. E.g., in this case, given a new cat/dog image, predict which of the two clusters it belongs to.

Can do it by assigning the image to the cluster with closer centroid

Slide Credit: Piyush Rai


Environment
Agent
1 Observe
2 Select action using policy
Ouch!
-50 points
3 Action!
4 Get reward or penalty
= bad!
...
Next time avoid it.
5 Update policy (learning step)
6 Iterate until an optimal policy is found

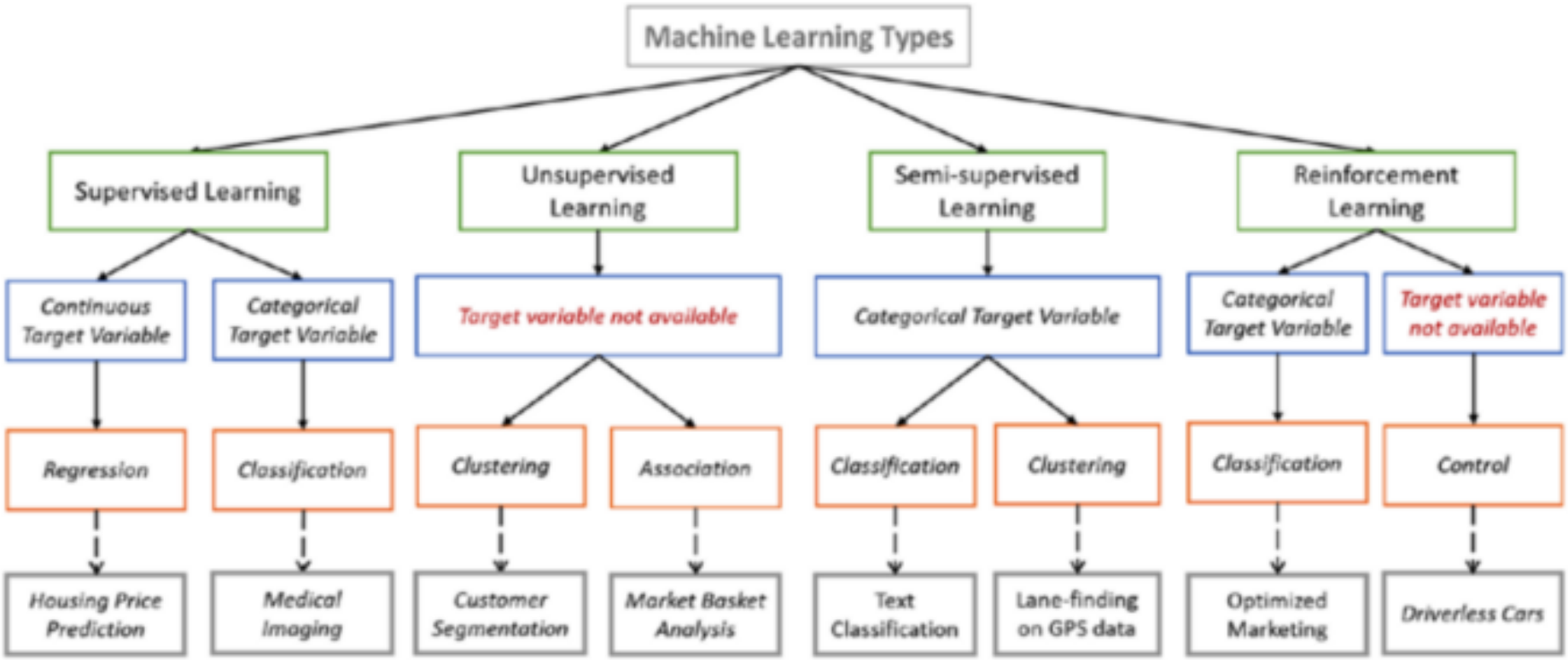


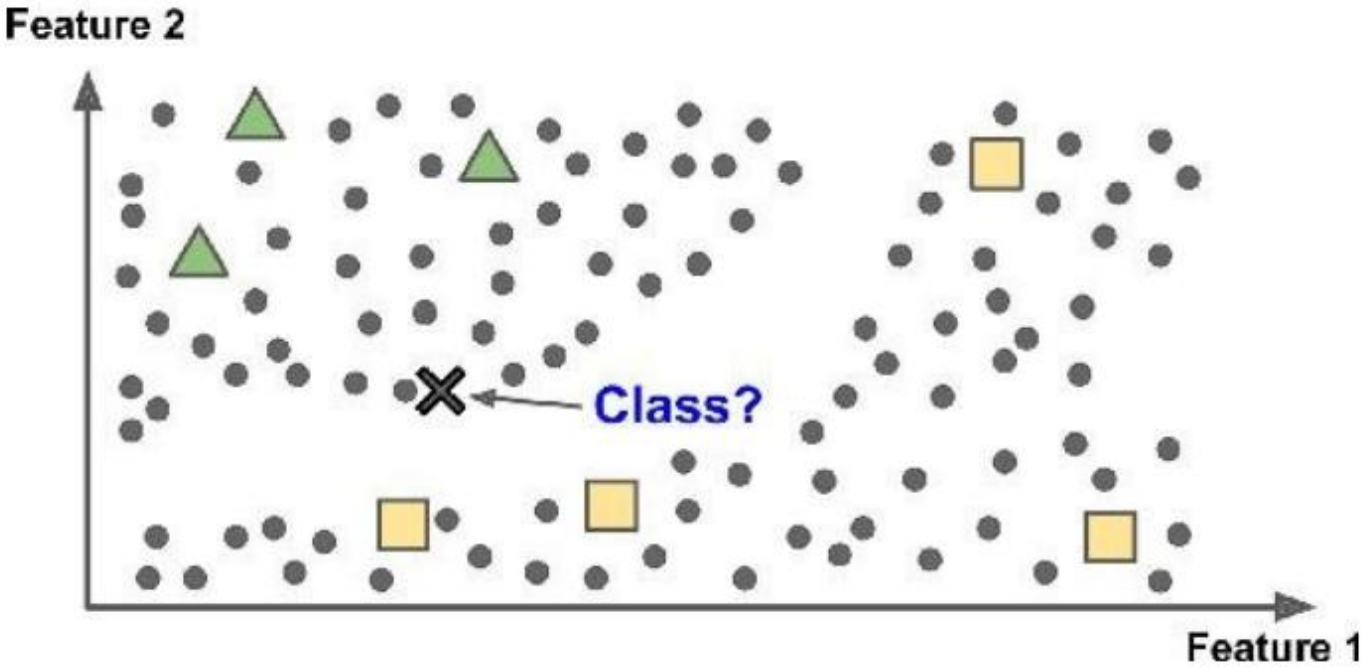
Image courtesy: https://towardsdatascience.com/types-of-machine-learning-algorithms-you-should-know-953a08248861

Criteria	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Definition	The machine learns by using labeled data	The machine is trained on unlabeled data without any guidance	An agent interacts with its environment by performing actions & learning from errors or rewards
Type of problems	Regression & classification	Association & clustering	Reward-based
Type of data	Labeled data	Unlabeled data	No predefined data
Training	External supervision	No supervision	No supervision
Approach	Maps the labeled inputs to the known outputs	Understands patterns & discovers the output	Follows the trial-and-error method

Semi supervised Learning

Partially labelled data – some labelled data and a lot of unlabelled data

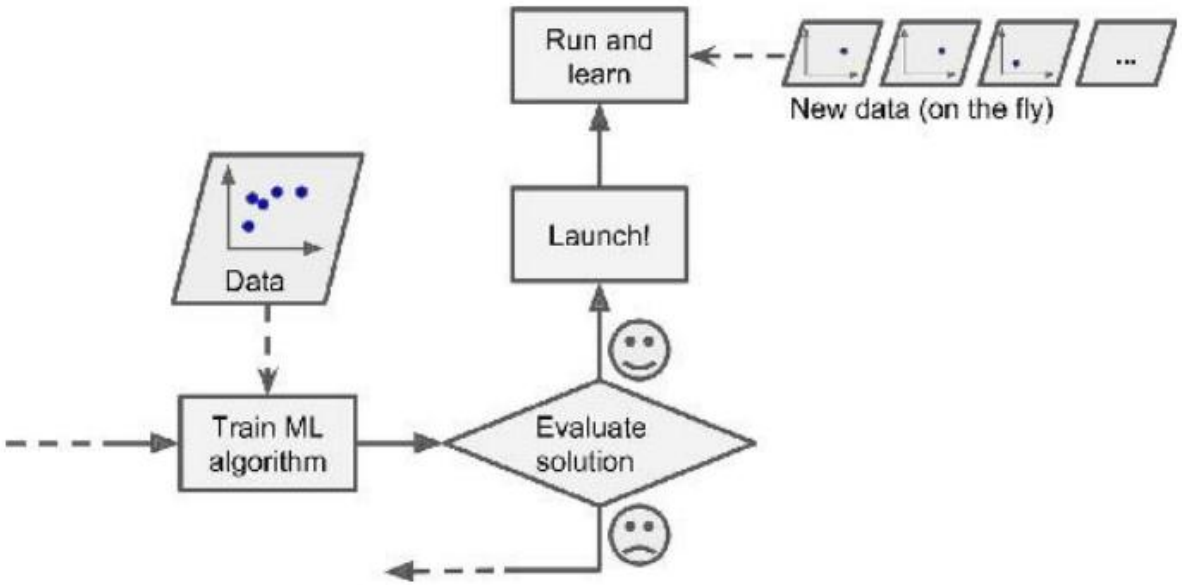
- Combines unsupervised and supervised learning algorithms
- Photo hosting service, e.g., google photos



Introduction to Machine Learning

Types: Based on how training data is used

- Batch learning: Uses all available data at a time during training
- Mini Batch learning: Uses a subset of available at a time during training
- Online (incremental) learning: Uses single training data instance at a time during training



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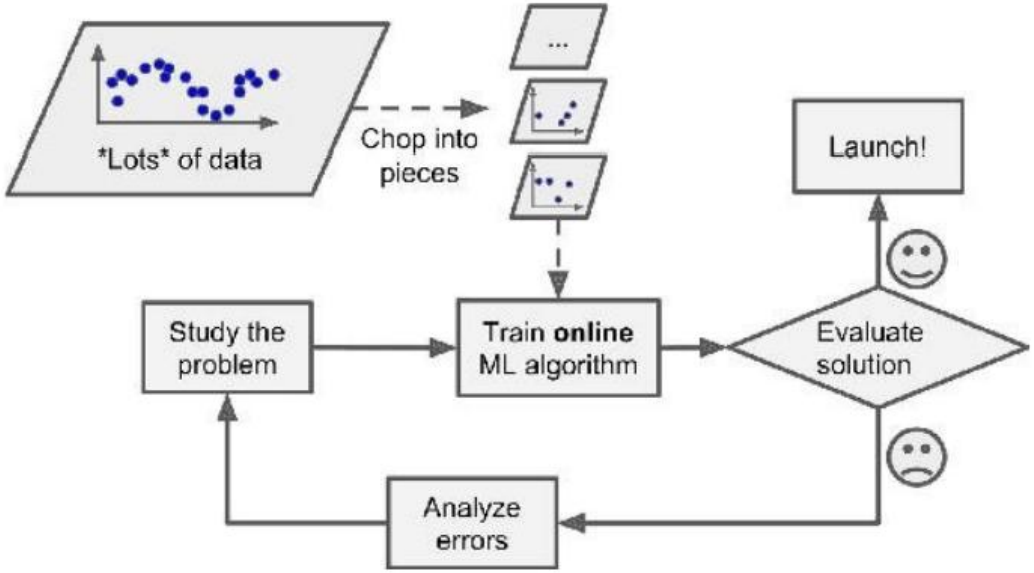
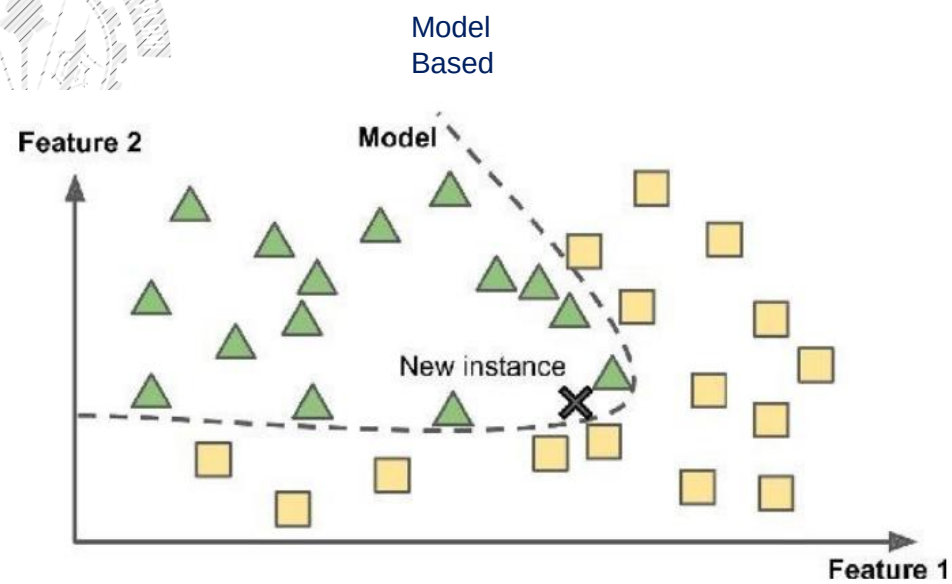


Introduction to Machine Learning

Types: Based on how training data is used

- Instance Based Learning: Compare new data points to known data points
- Model Based learning : Detect patterns in the training data and build a predictive model



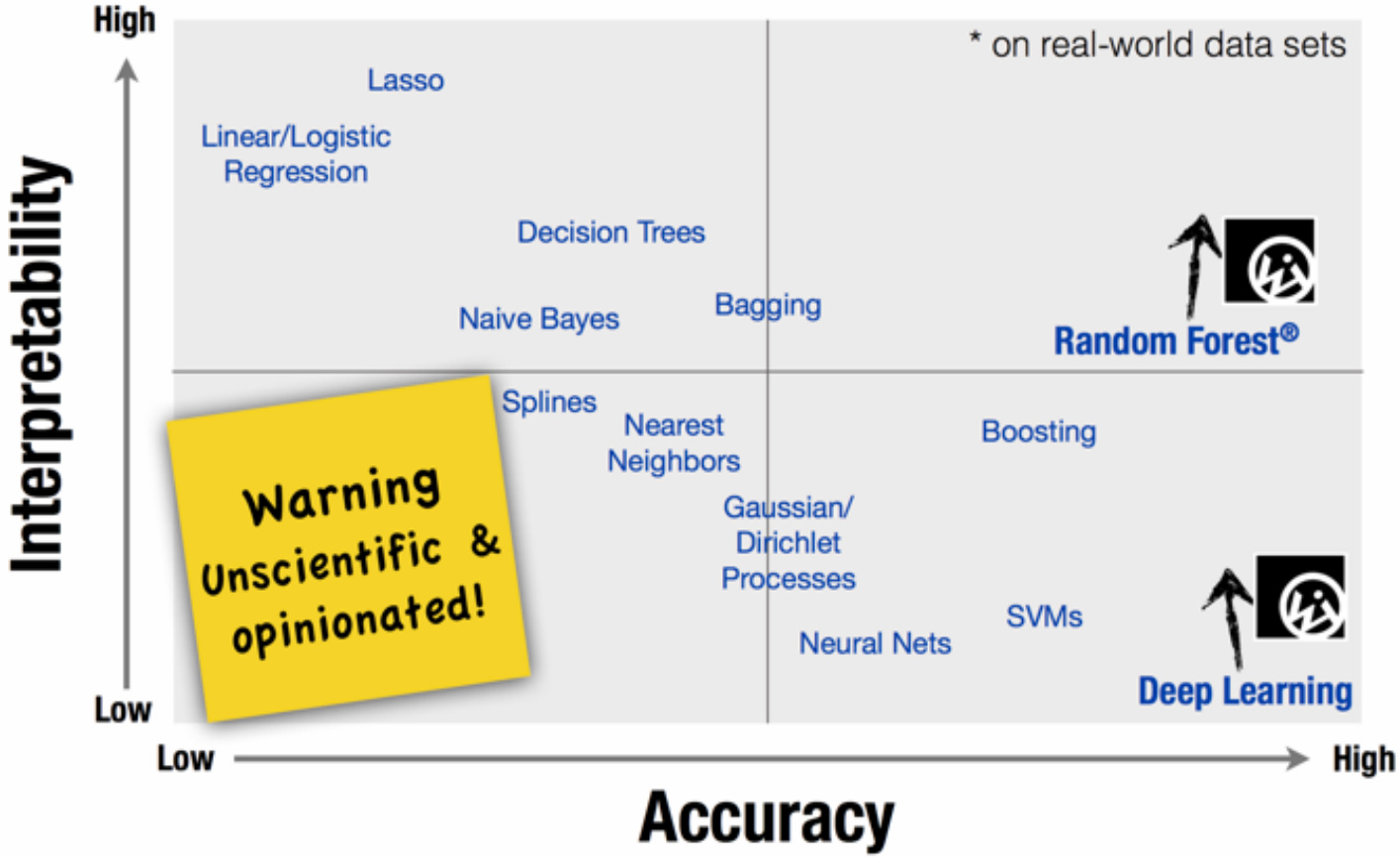
Open source ML programming tools

	Platform		Algorithms or Features
Scikit Learn	Linux, Mac OS, Windows	Python, C, C++	Classification, Regression, Clustering Preprocessing, Model Selection Dimensionality reduction.
PyTorch	Linux, Mac OS, Windows	Python, C++	Autograd Module, Optimization Module NN Module
TensorFlow	Linux, Mac OS, Windows	Python, C++	Provides a library for dataflow programming.
Weka	Linux, Mac OS, Windows	Java	Data preparation, Classification Regression, Clustering, Visualization Association rules mining

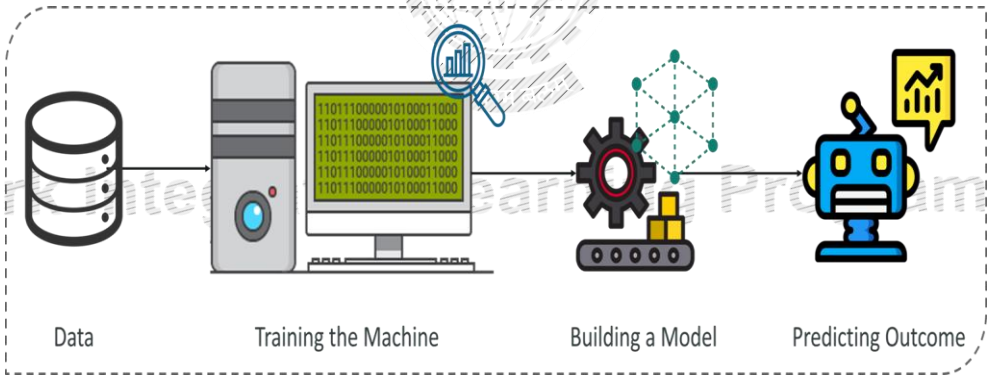
Open source ML programming tools

Colab	Cloud Service	-	Supports libraries of PyTorch, Keras, TensorFlow, and OpenCV
Apache Mahout	Cross-platform	Java Scala	Preprocessors,Regression Clustering, Recommenders Distributed Linear Algebra.
Accors.Net	Cross-platform	C#	Classification, Regression, Distribution Clustering, Hypothesis Tests & Kernel Methods, Image, Audio & Signal & Vision
Shogun	Windows Linux, UNIX Mac OS	C++	Regression, Classification, Clustering Support vector machines. Dimensionality reduction, Online learning etc.
Keras.io	Cross-platform	Python	API for neural networks

ML Algorithmic Trade-Off



ML workflow


Data
Training the Machine
Building a Model
Predicting Outcome

ML workflow

1. Should I use ML on this problem?
   - Is there a pattern to detect?
   - Can I solve it analytically?
   - Do I have data?
2. Gather and organize data.
3. Preprocessing, cleaning, visualizing.
4. Choosing a model, loss, regularization, ...
5. Optimization
6. Hyper parameter search.
7. Analyze performance and mistakes, and iterate back to step 5 (or 3)

Demo Case Study

: Car Price prediction based on Mileage

X Mileage	Y Car Price	H0	Error L1
9.8	10.48	9	2
9.12	1.75	8	7
9.5	6.95	9	3
10	2.51	9	7

- Define the Objective
- Data Gathering: survey, Past Purchase data
- Data Preprocessing
  - training set; test set
  - representation of input features; output
- Exploratory Data Analysis
- Choose form of model: linear regression
- System's performance evaluation: objective function
- Optimize performance by setting appropriate parameters: Optimization
- Evaluate on test set: generalization